

Bacterial Morphology - Factoid Sheet

Week 8 • Core Lecture 6 • Fisher deck primary • handout integrated

Master framework: morphology → surface structures → cell wall → Gram stain → division → spores.

Gram+ vs Gram-

Feature	Gram-positive	Gram-negative
Peptidoglycan	Thick	Thin
Membranes	1	2 (inner + outer)
Teichoic acids	Present	Absent
LPS	Absent	Outer membrane
Porins	-	Outer-membrane channels
Gram result	Purple	Pink/red

Gram stain: crystal violet → iodine (mordant) → acetone/alcohol (decolorizer) → safranin.

Technique trap: under-decolorize → organisms can look falsely Gram+. Over-decolorize → organisms can look falsely Gram-.

Peptidoglycan

Backbone Alternating NAG + NAM glycan chains.	Cross-link Peptide side chains; deck shows L-Ala, glutamate, lysine, D-Ala.
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Shape & Arrangement

Pattern	Recognition
Staphylococci	Cocci in grape-like clusters
Streptococci	Cocci in chains
Coccobacillus	Short plump rod / intermediate form
Corynebacterium	Club/dumbbell-shaped bacillus
Vibrio	Curved/comma-shaped bacillus
Spirillum	Rigid helix + polar flagella
Spirochete	Flexible helix + flagella below external sheath

Surface Structures

Structure	Deck function
Flagellum	Motility; filament-hook-basal body; flagellin
Fimbriae	Adherence
Sex/F pilus	Fertility/conjugation structure
Glycocalyx/capsule	Resists phagocytosis + attachment
Slime layer	Loose attachment + biofilm precursor + protection
Ribosome	30S + 50S = 70S

Handout clinical add-ons: 70S = 30S+50S and antibiotic target • Mycoplasma has no wall, uses host sterols, and ignores beta-lactams • PBPs/ transpeptidases cross-link peptidoglycan and are beta-lactam targets • LPS = O antigen + core + **Lipid A**; Lipid A/TLR4 drives inflammatory shock.

Division + Spores + Exam Traps

Bacterial Division

FtsZ = Z ring. Chromosome replicates/separates → Fts/divisome localizes → Z ring constricts → peptidoglycan/septum synthesis → daughter cells.

Endospore Formation

Step	Sequence in deck
1	DNA replication + axial filament
2	Asymmetric septum separates forespore / mother cell
3	Mother cell engulfs forespore → second membrane
4	Peptidoglycan layer develops between membranes
5	Calcium added to core; spore matures
6	Mother cell releases mature spore

Spore survival: extreme heat • lack of water • toxic chemicals • radiation.

Gram Morphology Examples

Appearance	Deck example
G+ cocci clusters	Staphylococcus
G+ cocci chains	Streptococcus
G+ cocci tetrads	Micrococcus
Thick G+ rods	Clostridium
Thin G+ rods	Listeria
Branched G+ rods	Actinomyces
G- diplococci	Neisseria gonorrhoeae
G- rods	E. coli, Klebsiella, Pseudomonas
Coccobacilli	Haemophilus influenzae

Assessment Answers Hidden in the Deck

Stem	Answer
Prokaryote but not eukaryote component	Peptidoglycan
Gram- feature absent from most Gram+	Porin
Why Gram+ retains crystal violet	Greater peptidoglycan
Mordant	Gram iodine
Gram+ negative polymer / cation movement	Teichoic acid
Most important Fts division protein	FtsZ
No-cell-wall pneumonia organism	Mycoplasma
Extrachromosomal resistance DNA	Plasmid
Peptidoglycan sugar	N-acetylglucosamine
Adherence projections made of pilins	Fimbriae
Membrane-anchored flagellum part	Basal body
Spirochete motility	Flagella beneath external sheath

Handout Vignette Connections

Clue	Answer
G- septic shock	Lipid A
Walking pneumonia + no wall	Beta-lactams ineffective
Capsule	Antiphagocytic
Reheated fried rice	Bacillus cereus endospore
E. coli-only envelope feature vs S. aureus	Outer membrane

Last 30 seconds: Gram+ thick PG/teichoic • Gram- outer membrane/LPS/porin • CV-I-alcohol-safranin • NAG/NAM • FtsZ • spores survive extremes.

Primary: current 78-slide Fisher deck. Official Week 8 LO sheet supplies the 6 objectives + assigned ScholarRx. Secondary enrichment: user-supplied Lecture 1: Bacterial Morphology and Structure handout.